VANISHING OF THE FIRST COHOMOLOGY OF MULTIPLES OF A RELATIVELY AMPLE BUNDLE ON A FLAT PROPER FAMILY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9518-9541 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9543-9619, the complete author proof (75 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Relative ampleness, cohomology and base change for flat proper morphisms, and finite local freeness criteria; the uniform-threshold argument is retained.

Source reference. Original lemma lemma-H1-O-picard-flat-proper: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9430–9451. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-diagonal-picard-flat-proper: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9199–9215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-algebra-lemma-lift-perfect-from-residue-field: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9619–9619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma perfect-lemma-affine-compare-bounded: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9619–9619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma perfect-lemma-flat-proper-perfect-direct-image-general: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9619–9619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma perfect-lemma-perfect-affine: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9619–9619. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider n commutative diagrams of schemes*

$$\begin{array}{ccc} X_i & \longrightarrow & X \\ & {\scriptstyle f_i} \searrow \quad \swarrow {\scriptstyle f} & \\ & S & \end{array}$$

with $f_i : X_i \to S$ and $f : X \to S$ satisfying the hypotheses of Lemma lemma-diagonal-picard-flat-proper. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module and let $\mathcal{L}_i$ be the pullback to X_i. Let $Z \subset S$, resp. $Z_i \subset S$ be the locally closed subscheme constructed in Lemma lemma-diagonal-picard-flat-proper for $(f, \mathcal{L})$, resp. $(f_i, \mathcal{L}_i)$ so that $Z \subset \bigcap_{i=1,\ldots,n} Z_i$. If $s \in Z$ and

$$H^1(X_s, \mathcal{O}) \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} H^1(X_{i,s}, \mathcal{O})$$

is injective, then $Z \cap U = (\bigcap_{i=1,\ldots,n} Z_i) \cap U$ (scheme theoretic intersection) for some open neighbourhood U of s.

Proof. This lemma is a variant of Lemma lemma-H1-O-picard-flat-proper and we strongly urge the reader to read that proof first; this proof is basically a copy of that proof with minor modifications. It follows from the description of (scheme valued) points of Z and the Z_i that $Z \subset \bigcap_{i=1,\ldots,n} Z_i$ where we take the scheme theoretic intersection. Thus we may replace S by the scheme theoretic intersection $\bigcap_{i=1,\ldots,n} Z_i$. After shrinking S to an affine open neighbourhood of s we may assume that $\mathcal{L}_i = \mathcal{O}_{X_i}$ for $i = 1, \ldots, n$. Let $E = Rf_*\mathcal{L}$ and $E_i = Rf_{i,*}\mathcal{L}_i = Rf_{i,*}\mathcal{O}_{X_i}$. These are perfect complexes whose formation commutes with arbitrary change of base (Derived Categories of Schemes, Lemma perfect-lemma-flat-proper-perfect-direct-image-general). In particular we see that

$$E \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s) = R\Gamma(X_s, \mathcal{L}_s) = R\Gamma(X_s, \mathcal{O}_{X_s})$$

The second equality because $s \in Z$. Set $h_j = \dim_{\kappa(s)} H^j(X_s, \mathcal{O}_{X_s})$. After shrinking S we can represent E by a complex

$$\mathcal{O}_S \to \mathcal{O}_S^{\oplus h_1} \to \mathcal{O}_S^{\oplus h_2} \to \ldots$$

see More on Algebra, Lemma more-algebra-lemma-lift-perfect-from-residue-field (strictly speaking this also uses Derived Categories of Schemes, Lemmas perfect-lemma-affine-compare-bounded and perfect-lemma-perfect-affine). Similarly, we may assume E_i is represented by a complex

$$\mathcal{O}_S \to \mathcal{O}_S^{\oplus h_{i,1}} \to \mathcal{O}_S^{\oplus h_{i,2}} \to \ldots$$

where $h_{i,j} = \dim_{\kappa(s)} H^j(X_{i,s}, \mathcal{O}_{X_{i,s}})$. By functoriality of cohomology we have a map

$$E \longrightarrow E_i$$

in $D(\mathcal{O}_S)$ whose formation commutes with change of base. Since the complex representing E is a finite complex of finite free modules and since S is affine, we

can choose a map of complexes

$$\begin{array}{ccccccc} \mathcal{O}_S & \xrightarrow[d]{} & \mathcal{O}_S^{\oplus h_1} & \longrightarrow & \mathcal{O}_S^{\oplus h_2} & \longrightarrow & \ldots \\ \downarrow{\scriptstyle a_i} & & \downarrow{\scriptstyle b_i} & & \downarrow{\scriptstyle c_i} & & \\ \mathcal{O}_S & \xrightarrow{d_i} & \mathcal{O}_S^{\oplus h_{i,1}} & \longrightarrow & \mathcal{O}_S^{\oplus h_{i,2}} & \longrightarrow & \ldots \end{array}$$

representing the given map $E \to E_i$. Since $s \in Z$ we see that the trivializing section of $\mathcal{L}_s$ pulls back to a trivializing section of $\mathcal{L}_{i,s} = \mathcal{O}_{X_{i,s}}$. Thus $a_i \otimes \kappa(s)$ is an isomorphism, hence after shrinking S we see that a_i is an isomorphism. Finally, we use the hypothesis that $H^1(X_s, \mathcal{O}) \to \bigoplus_{i=1,\ldots,n} H^1(X_{i,s}, \mathcal{O})$ is injective, to see that there exists a $h_1 \times h_1$ minor of the matrix defining $\oplus b_i$ which maps to a nonzero element in $\kappa(s)$. Hence after shrinking S we may assume that $(b_1, \ldots, b_n) : \mathcal{O}_S^{h_1} \to \bigoplus_{i=1,\ldots,n} \mathcal{O}_S^{h_{i,1}}$ is injective. However, since $\mathcal{L}_i = \mathcal{O}_{X_i}$ we see that $d_i = 0$ for $i = 1, \ldots n$. It follows that $d = 0$ because $(b_1, \ldots, b_n) \circ d = (\oplus d_i) \circ (a_1, \ldots, a_n)$. In this way we see that the trivializing section of $\mathcal{L}_s$ lifts to a section of $\mathcal{L}$ over X. A straightforward topological argument (omitted) shows that this means that $\mathcal{L}$ is trivial after possibly shrinking S a bit further. $\square$